Anisotropic dielectric image charges

This set of notes concerns the paper Screening of a point charge by an anisotropic medium: Anamorphoses in the method of images (https://pubs.aip.org/aapt/ajp/article-pdf/69/5/557/7529693/557_1_online.pdf) by Eugene Mele. We consider a point charge, Q, located a distance s above an anisotropic dielectric with $\varepsilon_x = \varepsilon_y = \varepsilon_1$ and $\varepsilon_z = \varepsilon_2$. Therefore, since we must have $\nabla \cdot \mathbf{D} = \rho_f$, and $\mathbf{D} = \varepsilon_0 \tilde{\varepsilon} \mathbf{E} = -\varepsilon_0 \tilde{\varepsilon} \nabla \Phi$, the equation for the total potential, Φ is given by: $-\varepsilon_0 \nabla \cdot \left(\tilde{\varepsilon} \nabla \Phi\right) = \rho_f$.

We solve this problem in four appendices. In **Appendix 1**, we write the scalar potential of a point charge in free space in cylindrical coordinates in terms of a linear combination of Bessel functions. In **Appendix 2**, we solve for the potential in the case of an isotropic dielectric semi-infinite half space. In **Appendix 3**, we solve for the potential in the case of a charge above a semi-infinite anisotropic dielectric.

Appendix 1: Scalar potential for a point charge in cylindrical coordinates

$$\nabla^2 \phi = -Q \delta(\mathbf{r}) / \varepsilon_0$$

We write the solution (given by **Equation 3** in the **Mele** paper but written in SI units here) to this equation as (at this point, the following is an *ansatz*, but we verify it below; we also show a different derivation at the end of this document):

$$\frac{Q}{4\pi\varepsilon_0} \int_0^\infty J_0(k\rho) e^{-k|z|} \mathrm{d}k,$$

where J_0 is the zeroth Bessel function of the first kind. Note that each component of this sum satisfies Laplace's equation, which we verify below in cylindrical coordinates:

$$\frac{1}{\rho}\partial_\rho(\rho\partial_\rho J_0(k\rho)) + k^2 J_0(k\rho) = \partial_\rho^2 J_0(k\rho) + \frac{1}{\rho}\partial_\rho J_0(k\rho) + k^2 J_0(k\rho) = \\ \frac{1}{\rho^2}\left[(k\rho)^2 J_0''(k\rho) + k\rho J_0'(k\rho) + (k\rho)^2 J_0(k\rho)\right] = 0,$$

where the last equality follows from Bessel's equation (for $m = 0$). Now we just need to satisfy the delta function at the origin. We note, in passing, that one may express the three dimensional delta function as $\delta(\mathbf{r}) = \delta(\rho)\delta(z)/2\pi\rho$ (which simply follows from $\int(\delta(\rho)\delta(z)/2\pi\rho)(2\pi\rho)\mathrm{d}\rho = 1$. We put a cylinder around the charge and calculate the flux of the electric field, which is given by:

$$\mathbf{E}(\rho, z) = -\frac{Q}{4\pi\varepsilon_0}\int_0^\infty k\Big[J_0'(k\rho)e_\rho \mp J_0(k\rho)e_z\Big]e^{-k|z|}\mathrm{d}k,$$

where the $(-, +)$ signs refer to the regions $z > 0, z < 0$, respectively. We then write the flux of the electric field as:

$$-\frac{Q}{4\pi\varepsilon_0}\int_0^\infty 2k\pi\rho J_0'(k\rho)\left[\int_0^z e^{-kz'}\mathrm{d}z' + \int_{-z}^0 e^{kz'}\mathrm{d}z'\right]\mathrm{d}k + \\ \frac{Q}{4\pi\varepsilon_0}\int_0^\infty 4\pi k e^{-k|z|}\int_0^\rho \rho' J_0(k\rho')\mathrm{d}\rho'\mathrm{d}k \\ = \int_0^\infty \frac{Q}{4\pi\varepsilon_0}\left[(4\pi\rho)J_0'(k\rho)\Big[e^{-kz} - 1\Big] + e^{-kz}\int_0^\rho 4\pi k\rho' J_0(k\rho')\mathrm{d}\rho'\right]\mathrm{d}k,$$

where the first term corresponds to the flux on the circular side of the cylinder and the second term corresponds to the flux through the caps of the cylinder. In order to evaluate the second term, we note that $J_0'(x) = -J_1(x)$, which means we may write:

$$\Big[xJ_1(x)\Big]' = J_1(x) + xJ_1'(x) = J_1(x) - x\left[\frac{1}{x}J_1(x) - J_0(x)\right] = xJ_0(x),$$

where, in the above, we used the fact that

$$J_1'(x) = (1/2)\Big[J_0(x) - J_2(x)\Big] = (1/2)\Big[J_0(x) - \Big[(2/x)J_1(x) - J_0(x)\Big]\Big] = J_0(x) - (1/x)J_1(x)$$

Using the above identity, we may then write (noting that $xJ_0(x)$ tends to zero at the origin, since $J_0(x) \to 1$ at the origin):

$$\int_0^x x' J_0(x') dx' = xJ_1(x)$$

We may therefore express the second term of the electric field's flux as (for each k) $(Q/\varepsilon_0)\rho J_1(k\rho)e^{-kz}$ and we may write the first term as $-(Q/\varepsilon_0)\rho J_1(k\rho)\Big[e^{-kz} - 1\Big]$, which means that the total flux is given by:

$$(Q/\varepsilon_0) \int_0^\infty \rho J_1(k\rho)\mathrm{d}k$$

To evaluate this, let's compute the integral:

$$\int_0^\infty J_1(k\rho)dk = (1/\rho)\int_0^\infty J_1(u)du = (1/\rho)J_0(0) = 1/\rho$$

Therefore, we have that the total flux is given by: Q/ε_0, as required.

Appendix 2: Solution for an isotropic dielectric

From the above, we know that the potential of a point charge in cylindrical coordinates a distance s above the dielectric is:

$$\Phi_{\text{point charge}}(\rho, z) = \frac{q}{4\pi\varepsilon_0}\int_k J_0(k\rho)e^{-k|z-s|}$$

Our goal now is to calculate the total potential everywhere in space, while obeying boundary conditions. The potential has to piecewise obey Laplace's equation and has to be azimuthally symmetric. In addition, the potential has to vanish asymptotically. This means that the total potential above the dielectric may be written as:

$$\Phi(\rho, z > 0) = \frac{q}{4\pi\varepsilon_0} \int_0^\infty J_0(k\rho)(e^{-k|z-s|} + \alpha(k)e^{-kz})\mathrm{d}k$$

And the total potential in the dielectric may be written as

$$\Phi(\rho, z < 0) = \frac{q}{4\pi\varepsilon_0} \int_0^\infty J_0(k\rho)(e^{-k|z-s|} + \beta(k)e^{kz})\mathrm{d}k$$

From the requirement that the potential match at $z = 0$ (equivalently that the tangential component of the electric field match at the boundary), we have that $\alpha(k) = \beta(k)$. In addition, we require the the normal component of the displacement field to be the same across the boundary. This gives us the condition:

$$\begin{aligned} ke^{-ks} - k\alpha(k) = \varepsilon\left[ke^{-ks} + \alpha(k)k\right] \rightarrow \alpha(k)k\left[\varepsilon + 1\right] = \left[1 - \varepsilon\right]ke^{-ks} \\ \rightarrow \alpha(k) = -\frac{\varepsilon - 1}{\varepsilon + 1}e^{-ks} \end{aligned}$$

Note that therefore, the total potential above the dielectric is given by:

$$\Phi(\rho, z > 0) = \frac{q}{4\pi\varepsilon_0} \int_0^\infty J_0(k\rho)\left[e^{-k|z-s|} - \frac{\varepsilon - 1}{\varepsilon + 1}e^{-k|z+s|}\right]\mathrm{d}k$$

This very obviously has the interpretation of the potential from the real charge added to the potential of an image charge $-q(\varepsilon - 1)/(\varepsilon + 1)$ located at $z = -s$

The potential in the dielectric is given by:

$$\Phi(\rho, z < 0) = \frac{q}{4\pi\varepsilon_0} \int_0^\infty J_0(k\rho)\left[e^{-k|z-s|} - \frac{\varepsilon - 1}{\varepsilon + 1}e^{-k|z-s|}\right]\mathrm{d}k$$

This is the potential of a point charge at $z = s$ with charge, $2q/(\varepsilon + 1)$

Appendix 3: Solution for an anisotropic dielectric (with charge in the vacuum infinite half-space)

For an anisotropic dielectric, we no longer have that the potential satisfies the Laplace equation. What we have instead is that $\nabla \cdot (\varepsilon \nabla \Phi) = 0$ in the dielectric and $\nabla^2 \Phi = 0$ in the $z > 0$ vacuum region. We may rewrite the above as:

$$\varepsilon_1 (1/\rho) \partial_\rho (\rho \partial_\rho \Phi) + \varepsilon_2 \partial_z^2 \Phi = 0$$

We may write this as Laplace's equation with the z component rescaled by $(\varepsilon_1/\varepsilon_2)^{1/2}$. I.e., we define $z' = (\varepsilon_1/\varepsilon_2)^{1/2} z$ and have that

$$(1/\rho) \partial_\rho (\rho \partial_\rho \Phi) + \partial_{z'}^2 \Phi = 0$$

The solutions to this equation are linear combinations of terms of the form

$$\Phi_k(\rho, z) = J_0(k\rho) e^{\pm k z'} \to \Phi(\rho, z) = \int_0^\infty \tilde{G}(k) J_0(k\rho) e^{\pm kz/\gamma} \mathrm{d}k$$

To put the above in the form of the Mele paper, we replace $k \to k'$ and subsequently take $k = k'/\gamma$, which means that $G(q)$ (defined in the Mele paper) is given by $G(q) = \gamma \tilde{G}(q)$.

Therefore, we write the potential in the vacuum region as:

$$\Phi(\rho, z > 0) = \frac{q}{4\pi\varepsilon_0} \int_0^\infty J_0(k\rho) \left[e^{-k|z-s|} + \alpha(k) e^{-kz} \right] \mathrm{d}k$$

And we write the potential in the dielectric as:

$$\Phi(k, z < 0) = \frac{q}{4\pi\varepsilon_0} \int_0^\infty \beta(k) J_0(k\gamma\rho) e^{kz} \mathrm{d}k$$

From this, we see that $\beta(k) = \gamma \left[e^{-k\gamma s} + \alpha(k\gamma) \right]$ as well as $k\gamma^2 (e^{-k\gamma s} - \alpha(k\gamma)) = \varepsilon_2 k \beta(k)$

Putting these two equations together, we obtain:

$$\gamma(e^{-k\gamma s} - \alpha(k\gamma)) = \varepsilon_2 \left[e^{-k\gamma s} + \alpha(k\gamma) \right] \to \alpha(k\gamma) \left[\varepsilon_2 + \gamma \right] = (\gamma - \varepsilon_2) e^{-k\gamma s}$$

Therefore, this gives us:

$$\alpha(k) = \left[\frac{(\varepsilon_2/\varepsilon_1)^{1/2} - \varepsilon_2}{(\varepsilon_2/\varepsilon_1)^{1/2} + \varepsilon_2}\right] e^{-ks} = \left[\frac{1 - \sqrt{\varepsilon_1 \varepsilon_2}}{1 + \sqrt{\varepsilon_1 \varepsilon_2}}\right] e^{-ks}$$

Thus, the potential in the vacuum region is given by:

$$\Phi(\rho, z > 0) = \frac{q}{4\pi\varepsilon_0} \int_k J_0(k\rho) \left[e^{-k|z-s|} - \frac{\sqrt{\varepsilon_1\varepsilon_2} - 1}{\sqrt{\varepsilon_1\varepsilon_2} + 1} e^{-k|z+s|} \right] \mathrm{d}k$$

Therefore, the image charge is again located at $z = -s$ and has a charge of (in accordance with **Equation 20** of the **Mele** paper):

$$-q\left[\frac{\sqrt{\varepsilon_1\varepsilon_2} - 1}{\sqrt{\varepsilon_1\varepsilon_2} + 1}\right]$$

Giving a total potential of:

$$\frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(\rho^2 + (z-s)^2)} - \frac{\sqrt{\varepsilon_1\varepsilon_2} - 1}{\sqrt{\varepsilon_1\varepsilon_2} + 1} \frac{1}{(\rho^2 + (z+s)^2)}\right]$$

The potential in the dielectric is given by:

$$\frac{q\gamma}{4\pi\varepsilon_0} \int_k J_0(k\gamma\rho) \left[\frac{2}{1 + \sqrt{\varepsilon_1\varepsilon_2}}\right] e^{-k(\gamma s - z)} = \frac{q}{4\pi\varepsilon_0} \int_k J_0(k\rho) \left[\frac{2}{1 + \sqrt{\varepsilon_1\varepsilon_2}}\right] e^{-k|s - z/\gamma|)} = \frac{q}{4\pi\varepsilon_0}\left[\frac{2}{1 + \sqrt{\varepsilon_1\varepsilon_2}}\right] \frac{1}{(\rho^2 + (s - z/\gamma)^2)^{1/2}},$$

which is **Equation 21** in the **Mele** paper.

Appendix 4: Charge in the anisotropic dielectric

If the charge is in the dielectric, we take the potential above the surface to be given by:

$$\Phi(\rho, z>0) = \frac{q}{4\pi\varepsilon_0}\int_0^\infty \alpha(k) J_0(k\rho) e^{-kz}\mathrm{d}k$$

In the dielectric, the equation to be satisfied is given by:

$$\varepsilon_1(1/\rho)\partial_\rho(\rho\partial_\rho\Phi) + \varepsilon_2\partial_z^2\Phi = -Q\delta(\mathbf{r}-\mathbf{r}_0)/\varepsilon_0$$

So we need to make sure that the flux of $\mathbf{D}$ on a cylinder encompasing the charge gives the proper delta function.

We write the potential below the plane as:

$$\Phi(\rho, z<0) = \frac{q}{4\pi\varepsilon_0}\int_0^\infty J_0(k\rho)\left[\beta(k)e^{-k|s+z|/\gamma} + \gamma(k)e^{kz/\gamma}\right]\mathrm{d}k$$

The second component is a homogeneous solution to Laplace's equation as is the first component everywhere except at $z=-s$. We first try to find $\beta(k)$. To do so, we recenter at $s=0$ (imagining a dielectric of infinite extent). And we find the displacement field:

$$-\frac{q}{4\pi}\int_0^\infty \beta(k)\left[\varepsilon_1 J_0'(k\rho)e_\rho \mp (\varepsilon_2/\gamma)kJ_0(k\rho)e_z\right]e^{-k|z|/\gamma}\mathrm{d}k$$

The flux of the displacement field is then (for each k component):

$$\frac{q}{4\pi}\beta(k)\gamma\varepsilon_1(4\pi)\rho J_0'(k\rho)\left[e^{-kz/\gamma}-1\right] + \frac{q}{4\pi}\beta(k)(\varepsilon_2/\gamma)(4\pi)e^{-kz/\gamma}\int_0^\rho k\rho' J_0(k\rho')\mathrm{d}\rho'$$

We re-express this as:

$$q\beta(k)\rho\left[\gamma\varepsilon_1 J_1(k\rho)\left[1-e^{-kz/\gamma}\right] + (\varepsilon_2/\gamma)e^{-kz/\gamma}J_1(k\rho)\right]$$

Note that $\gamma\varepsilon_1 = \varepsilon_2/\gamma = (\varepsilon_1\varepsilon_2)^{1/2}$, which allows us to simplify our expression for the flux as:

$$(\varepsilon_1\varepsilon_2)^{1/2} q\int_0^\infty \beta(k)\rho J_1(k\rho)\mathrm{d}k$$

Therefore, we have that $\beta(k) = 1/(\varepsilon_1\varepsilon_2)^{1/2}$

At this point, we impose the boundary conditions, which give us:

$$\frac{1}{\sqrt{\varepsilon_1\varepsilon_2}} e^{-ks/\gamma} + \gamma(k) = \alpha(k)$$
$$-\alpha(k) = \frac{\varepsilon_2}{\gamma}\gamma(k) - \frac{\varepsilon_2}{\gamma} e^{-ks/\gamma}\frac{1}{\sqrt{\varepsilon_1\varepsilon_2}} = \sqrt{\varepsilon_1\varepsilon_2}\gamma(k) - e^{-ks/\gamma}$$

We solve for $\gamma(k)$:

$$-\left[\frac{1}{\sqrt{\varepsilon_1\varepsilon_2}} e^{-ks/\gamma} + \gamma(k)\right] = \sqrt{\varepsilon_1\varepsilon_2}\gamma(k) - e^{-ks/\gamma} \to$$
$$(1+\sqrt{\varepsilon_1\varepsilon_2})\gamma(k) = e^{-ks/\gamma}\left[1 - \frac{1}{\sqrt{\varepsilon_1\varepsilon_2}}\right] \to \gamma(k) = \frac{e^{-ks/\gamma}}{\sqrt{\varepsilon_1\varepsilon_2}}\left[\frac{\sqrt{\varepsilon_1\varepsilon_2}-1}{\sqrt{\varepsilon_1\varepsilon_2}+1}\right],$$
$$\alpha(k) = e^{-ks/\gamma}\left[\frac{2}{\sqrt{\varepsilon_1\varepsilon_2}+1}\right]$$

This seems to indicate a sign error in **Equations 30, 31** in the **Mele** paper.

At this point, we write out the potentials in Cartesian coordinates. First, we note that above the plane, we have:

$$\Phi(\rho, z>0) = \frac{q}{4\pi\varepsilon_0}\int_0^\infty \alpha(k) J_0(k\rho) e^{-kz}\mathrm{d}k = \frac{q}{4\pi\varepsilon_0}\int_0^\infty e^{-k(z+s/\gamma)}\left[\frac{2}{\sqrt{\varepsilon_1\varepsilon_2}+1}\right]\mathrm{d}k$$

From the first appendix, we see clearly that this corresponds to the following potential:

$$\Phi(\rho, z>0) = \frac{q}{4\pi\varepsilon_0}\frac{1}{\sqrt{(z+s/\gamma)^2+x^2+y^2}}\frac{2}{\sqrt{\varepsilon_1\varepsilon_2}+1},$$

which is **Equation 32**. Furthermore, for the potential below the plane, we have:

$$\Phi(\rho, z<0) = \frac{q}{4\pi\varepsilon_0}\int_0^\infty J_0(k\rho)\left[\beta(k)e^{-k|s+z|/\gamma} + \gamma(k)e^{kz/\gamma}\right]\mathrm{d}k = \frac{q}{4\pi\varepsilon_0}\int J_0(k\rho)\left[\frac{1}{\sqrt{\varepsilon_1\varepsilon_2}}e^{-k|s+z|/\gamma} + \frac{e^{-k(s-z)/\gamma}}{\sqrt{\varepsilon_1\varepsilon_2}}\left[\frac{\sqrt{\varepsilon_1\varepsilon_2}-1}{\sqrt{\varepsilon_1\varepsilon_2}+1}\right]\right] = \frac{q}{4\pi\varepsilon_0\sqrt{\varepsilon_1\varepsilon_2}}\left[\frac{1}{\sqrt{x^2+y^2+(z+s)^2/\gamma^2}} + \frac{1}{\sqrt{x^2+y^2+(z-s)^2/\gamma^2}}\left[\frac{\sqrt{\varepsilon_1\varepsilon_2}-1}{\sqrt{\varepsilon_1\varepsilon_2}+1}\right]\right],$$

which is **Equation 33**.

Alternative derivation of Bessel function representation of Coulomb potential

We wish to find a solution to the equation:

$$\nabla^2\phi(\mathbf{r}) = -Q\delta(\mathbf{r})/\varepsilon_0$$

in cylindrical coordinates. We write this as a cylindrical Fourier decomposition:

$$\phi(\mathbf{r}) = \frac{1}{4\pi^2}\int\phi(\mathbf{q}, z)e^{i\mathbf{q}\cdot\rho}\mathrm{d}^2\mathbf{q},\ \text{where } (\partial_z^2 - |\mathbf{q}|^2)\phi(\mathbf{q}, z) = 0 \text{ everywhere except } \mathbf{r} = 0$$

From this, we obtain:

$$\phi(\mathbf{r}) = \frac{1}{4\pi^2}\int\phi(\mathbf{q})e^{i\mathbf{q}\cdot\rho}e^{-|\mathbf{q}|z}\mathrm{d}^2\mathbf{q} \to \partial_z\phi(\mathbf{r}|z=0+) - \partial_z\phi(\mathbf{r}|z=0-) = \frac{1}{4\pi^2}\int(2|\mathbf{q}|)\phi(\mathbf{q})e^{i\mathbf{q}\cdot\rho}\mathrm{d}^2\mathbf{q} = (Q/\varepsilon_0)\int\frac{1}{4\pi^2}e^{i\mathbf{q}\cdot\rho}\mathrm{d}^2\mathbf{q}$$

From this, we obtain $\phi(\mathbf{q}) = Q/2|\mathbf{q}|\varepsilon_0$, which gives us:

$$\phi(\mathbf{r}) = \frac{Q}{8\pi^2\varepsilon_0}\int\int e^{i|\mathbf{q}|\rho\cos(\theta)}\mathrm{d}\theta e^{-|\mathbf{q}|z}\mathrm{d}|\mathbf{q}| = \frac{Q}{4\pi\varepsilon_0}\int J_0(\rho|\mathbf{q}|)e^{-|\mathbf{q}|z}\mathrm{d}|\mathbf{q}|,$$

as desired.